MILITARY SPECIFICATION

PRIMERS, PERCUSSION, M31A2 AND M31B2 PARTS FOR

1. SCOPE

1.1 This specification covers the parts for percussion primers M31A2 and M31B2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1390

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DRAWINGS

ORDNANCE CORPS

8838012 - Cup, Battery
8838015 - Plug, Firing
8838080 - Head
8838081 - Head
8838089 - Body

PUBLICATIONS

ORDNANCE CORPS

IEL-7289906 - Index of Inspection Equipment Lists.
IEL-7289920 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer components.-The primer components shall comply with all requirements specified on Drawings (dwg.) 8838012, 8838015, 8838080, 8838081, and 8838089 as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test (applicable only to steel primer tubing).-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.4.1 (see 6.4).

3.4 Workmanship.-All parts shall be fabricated and finished in a through, workmanlike manner. They shall be clean and

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free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind

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his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 At the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.5) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of (25) twenty-five complete set of parts (i.e., (25) twenty-five each of every component which have been produced by the contractor or furnished by a supplier and which has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and

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submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production is accepted, the contractor is in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of inert components produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or interchangeability, as determined by the Government, shall not necessitate changing the lot number. Lots shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235, when applicable).

4.3.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Plug, firing (see dwg. 8838015).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Large diameter, min.	Gage	01001
102.	Total length, min.	Gage	01002
103.	Length and profile of point.	Gage	01003
104.	Length of pilot, min.	Gage	01004
Minor:	AQL 1.5 percent		
201.	Small diameter, min.	Gage	01005
202.	Large diameter, maximum (max.)	Gage	01006
203.	Length of pilot, max.	Gage	01007
204.	Evidence of poor workmanship (see 3.4)	Visual	01008

4.3.2.2 Head (see dwg. 8838080 and 8838081).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Pitch diameter of thread.	Gage	02001
102.	Major diameter of thread.	Gage	02002
103.	Diameter of body pilot, max.	Gage	02003
104.	Diameter of battery cup cavity.	Gage	02004
105.	Diameter of percussion element cavity.	Gage	02005
106.	Diameter under flange.	Gage	02006

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107.	Depth of percussion element cavity	Gage	02007
108.	Thickness of flange, max	Gage	02008
109.	Crack or split	Visual	02009
110.	Flash hole missing or obstructed	Visual	02010
Minor:	AQL 6.5 percent		
201.	Diameter of flange	Gage	02011
202.	Depth of battery cup cavity	Gage	02012
203.	Diameter of body pilot, min	Gage	02013
204.	Concentricity between percussion element cavity and flash hole	Gage	02014
205.	Concentricity of percussion element cavity with diameter under flange	Gage	02015
206.	Concentricity of flange with diameter under flange	Gage	02016
207.	Diameter of chamber at mouth of percussion element cavity	Gage	02017
208.	Diameter of flash hole	Gage	02018
209.	Length to shoulder above thread	Gage	02019
210.	Length of thread undercut, max	Gage	02020
211.	Length of threads including thread undercut	Gage	02021
212.	Total length	Gage	02022
213.	Depth of flash cavity	Gage	02023
214.	Diameter and profile of flash cavity	Gage	02024
215.	Thread undercut missing	Visual	02025
216.	Radius missing	Visual	02026
217.	Finish improper	Visual	02027
218.	Evidence of poor workmanship (see 3.4)	Visual	02028

4.3.2.3 Body (see dwg. 8838089).

Categories	Defects	Method of Inspection	Code No.

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Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Flash hole missing	Gage	03001
2.	Crack or split	Visual	03002
Major:	AQL 1.0 percent		
101.	Pitch diameter of thread	Gage	03003
102.	Length of thread, min	Gage	03004
103.	Minor diameter of thread, max	Gage	03005
104.	Wall thickness, min	Gage	03006
105.	Thread stripped or otherwise damaged	Visual	03007
Minor:	AQL 4.0 percent		
201.	Length of thread, max	Gage	03008
202.	Location of first row of holes	Gage	03009
203.	Inside diameter min	Gage	03010
204.	Depth of counterbore, max	Gage	03011
205.	Diameter of counterbore	Gage	03012
206.	Total length, max	Gage	03013
207.	Plug gage fails to pass required depth of body	Gage	03014
208.	Body fails to pass through required ring gage	Gage	03015
209.	Concentricity of inside diameter with pitch diameter of thread	Gage	03016
210.	Finish improper	Visual	03017
211.	Protective coating damaged (base metal exposed)	Visual	03018
212.	Evidence of poor workmanship (see 3.4)	Visual	03019

4.3.2.4 Cup, battery (see dwg. 8838012).

Categories	Defects	Method of Inspection	Code No.

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Major:	AQL 0.65 percent		
101.	Outside diameter, min	Gage	04001
102.	Thickness through bottom, maximum (max)	Gage	04002
103.	Length	Gage	04003
Minor:	AQL 2.5 percent		
201.	Diameter of hole	Gage	04004
202.	Thickness through bottom, min	Gage	04005
203.	Inside diameter	Gage	04006
204.	Outside diameter, max	Gage	04007
205.	Flat missing from surface at open end	Visual	04008
206.	Evidence of poor workmanship (see 3.4)	Visual	04009

4.3.3 Testing

4.3.3.1 Hydrostatic test (see 3.3) Critical defect - Code No. 04010.-This test shall be conducted 100 percent. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom. The test shall be performed as specified in 4.4.1 using equipment specified in 4.3.4.

4.3.4 Inspection Equipment.-Index of Inspection Equipment Lists Number IEL 7289906 and 7289920 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

$$P = (1.8\ St) \text{ divided by } D$$

where:

P = test pressure in pounds per square inch (psi).
S = minimum specified yield strength, psi.
t = specified minimum wall thickness in inches.
D = specified maximum outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-P-10025.

5.2 Data cards.-Data cards shall be prepared for each lot in accordance with information specified in Standard MIL-STD-1167.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

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6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection

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equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Hydrostatic test.-If the primer tubing procured under this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

6.5 Submission of initial production samples.-Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND.

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Preparing Activity:
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